RESEARCH SEMINAR · LAKEHEAD UNIVERSITY

From Quantitative Imaging to Predictive Virtual Trials

Computational approaches toward precision medicine

Hamid Abdollahi, PhD
Medical physics · Molecular imaging · Theranostics · Digital twins

Imaging

Modelling

Personalized therapy

OVERVIEW

Outline

Introduction

- Cancer complexity and heterogeneity
- The cancer technology ecosystem
- Imaging the tumour phenotype
- Radiomics: workflows and applications

Research & methods

- Radiomics studies: diagnosis, response, toxicity
- Feature reliability and harmonization
- Digital twins and theranostics
- Predictive dosimetry

Research vision

- I Imaging biomarkers
- II AI-enhanced imaging
- III Digital twins for RPT
- IV Advanced therapies
- V Virtual trials

BEFORE WE START

Disclosure

Selected PhD and postdoctoral work that shaped my approach to precision medicine

No conflicts of interest to declare

Thanks to all collaborators who contributed to these works

Thank you to the search committee for this opportunity

PART I · 35 MINUTES

Research background and current work

Cancer complexity · imaging biomarkers · radiomics · digital twins · theranostics

Cancer is the leading cause of death in Canada

2 in 5
Canadians will develop cancer

1 in 4
Canadians will die from cancer

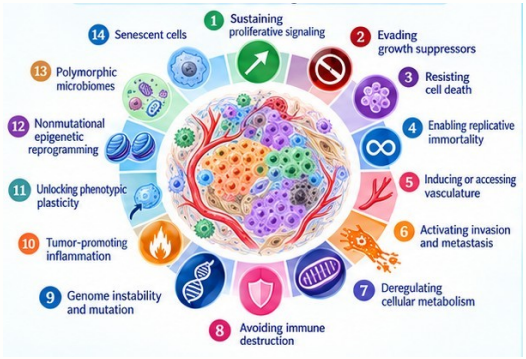
#1
cause of death in Canada

254,800
new diagnoses expected in 2025

87,400
cancer deaths expected in 2025

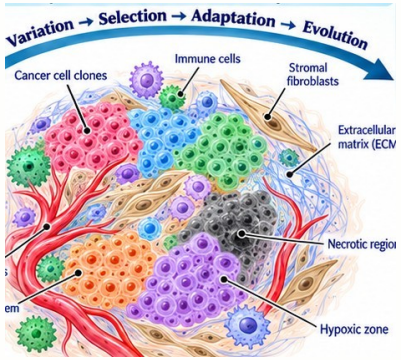
64%
five-year net survival

A complex, dynamic and heterogeneous disease



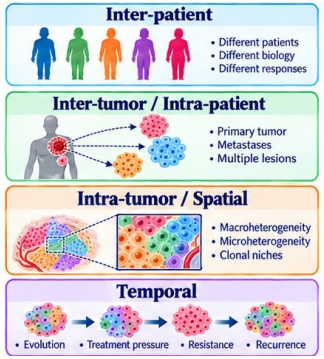
1 Complex biology

- 14 hallmarks and enabling characteristics
- Genetic and epigenetic alterations
- Shaped by the microenvironment



2 Dynamic ecosystem

- Cancer, immune, stromal cells and vessels
- Variation → selection → adaptation → evolution
- Hypoxia, immunity and treatment drive change



3 Heterogeneous

- Between patients
- Between lesions in one patient
- Within a tumour and over time

Clinical consequences: variable prognosis · differential treatment response · resistance · metastasis · recurrence

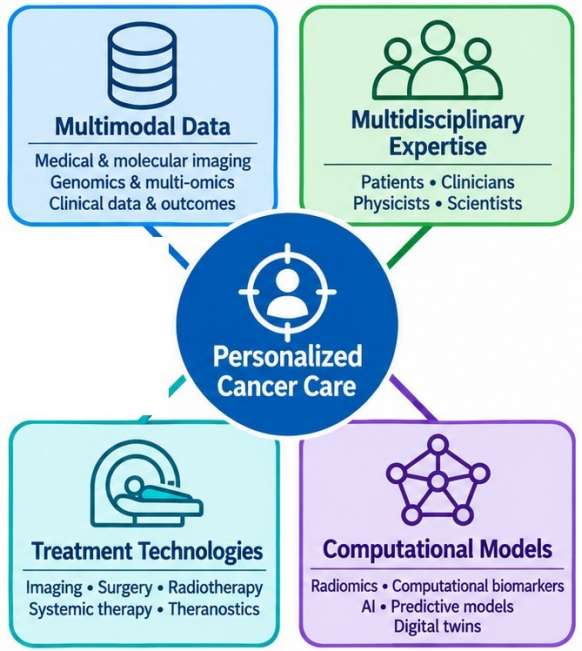
Hanahan & Weinberg, Cell 144, 2011 · Kashyap et al., Trends in Biotechnology, 2022

From the care continuum to personalized care



1 Care continuum

- Prevention and screening
- Diagnosis and stratification
- Treatment planning and delivery
- Response assessment
- Survivorship and monitoring

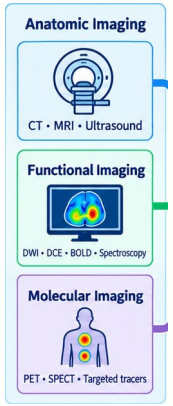


2 Personalized care

- Multimodal data: imaging, omics, outcomes
- Multidisciplinary expertise
- Treatment technologies
- Computational models: radiomics, AI, digital twins

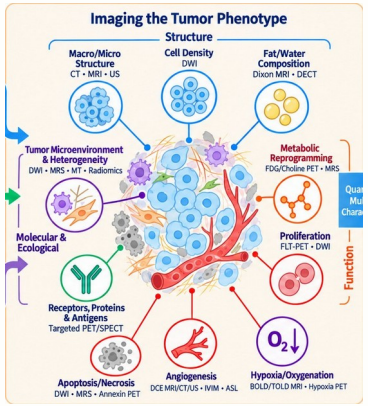
Multimodal · multiscale · multipurpose · multidisciplinary — evidence → decisions → outcomes → learning

Imaging the tumour phenotype



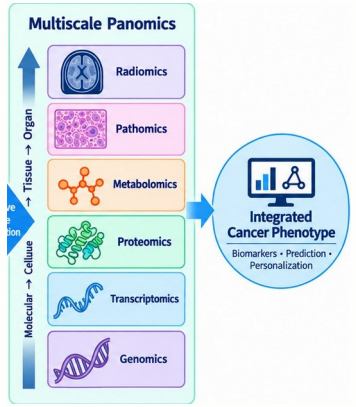
1 Imaging modalities

- Anatomic: CT, MRI, ultrasound
- Functional: DWI, DCE, BOLD, MRS
- Molecular: PET, SPECT, tracers



2 Tumour hallmarks we can image

- Structure: cell density, fat/water
- Function: metabolism, proliferation, hypoxia, angiogenesis
- Receptors, apoptosis, microenvironment



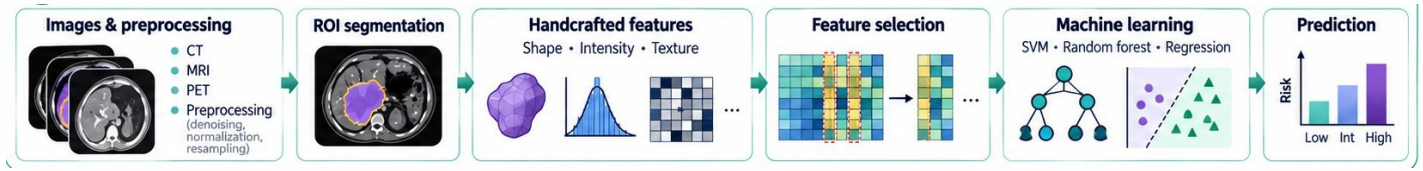
3 Multiscale panomics

- Radiomics, pathomics, metabolomics, proteomics, genomics
- → integrated cancer phenotype

Quantitative, multiscale imaging → biomarkers, prediction and personalization

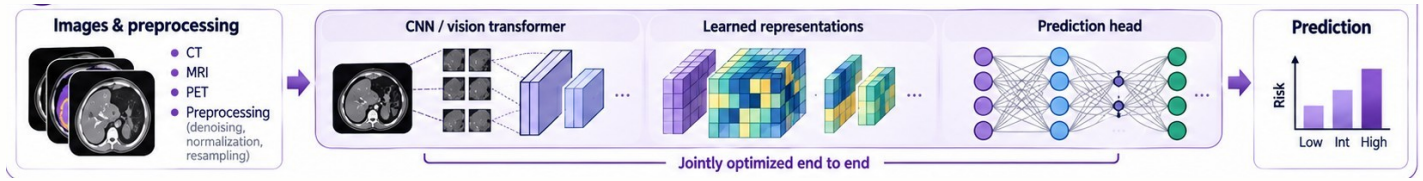
IMAGING BIOMARKER · RADIOMICS

Three radiomics workflows



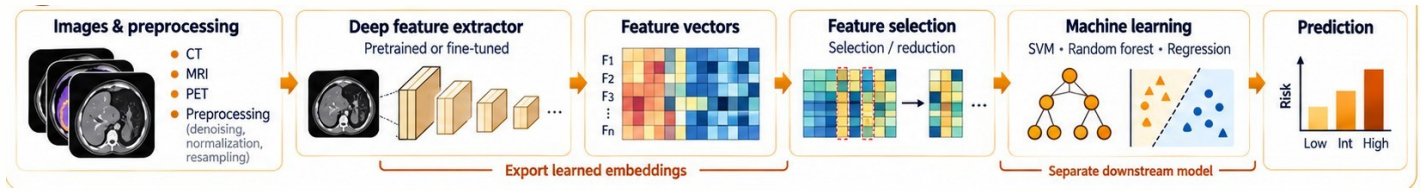
A Handcrafted radiomics

- ROI → shape, intensity, texture features
- Feature selection + machine learning



B End-to-end deep learning

- CNN / transformer learns the features
- Optimized end to end, image → risk



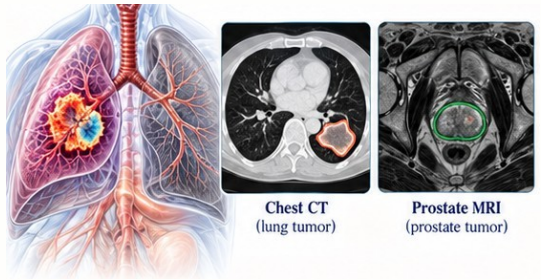
C Deep features + ML

- Pretrained network exports embeddings
- Separate ML model predicts risk

Applications: diagnosis · prognosis · treatment response — independent validation required

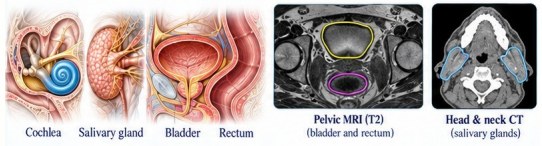
IMAGING BIOMARKER · RADIOMICS

Radiomics: from tumour to whole patient



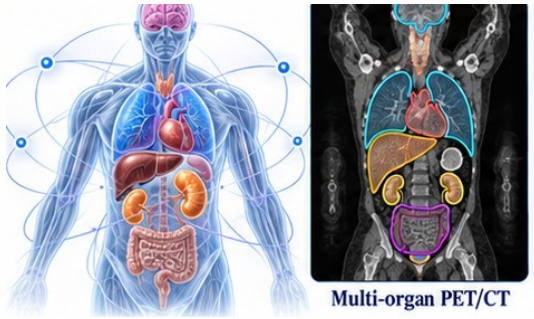
I Tumour phenotype

- Prostate, lung, brain, kidney
- Grade, molecular status, prognosis
- Treatment-response prediction



II Normal tissue

- Cochlea, salivary glands, rectum, bladder
- Changes after radiotherapy
- Toxicity-risk prediction



III Systemic, multi-organ

- Host beyond the tumour
- Endocrine-organ signatures
- PSMA-negative prostate cancer



IV Non-oncologic

- Myocardial infarction (cardiac MRI)
- COVID-19 lung disease (CT)
- Bone mineral loss (radiograph)

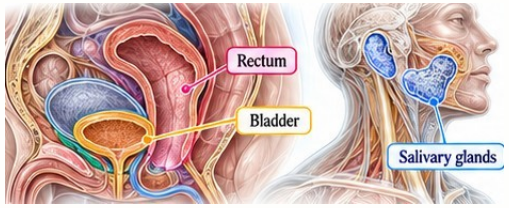
Cross-cutting: multimodal imaging · radiogenomics · longitudinal / delta radiomics

Applications of radiomics



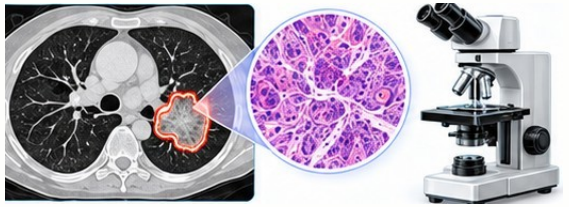
I RT response

- Predict tumour response to RT
- Identify resistance patterns



II Toxicity

- Normal-tissue injury risk
- Salivary glands, rectum, bladder



III Diagnosis

- Tumour type, grade, aggressiveness
- Support lesion characterization



IV Prognosis

- Progression and recurrence risk
- Stratify patients by outcome



V Radiogenomics

- Link features to tumour biology
- Molecular signatures



VI Monitoring

- Track change during and after therapy
- Delta radiomics

Beyond the tumour: systemic and non-cancer disease · needs reproducible features and external validation

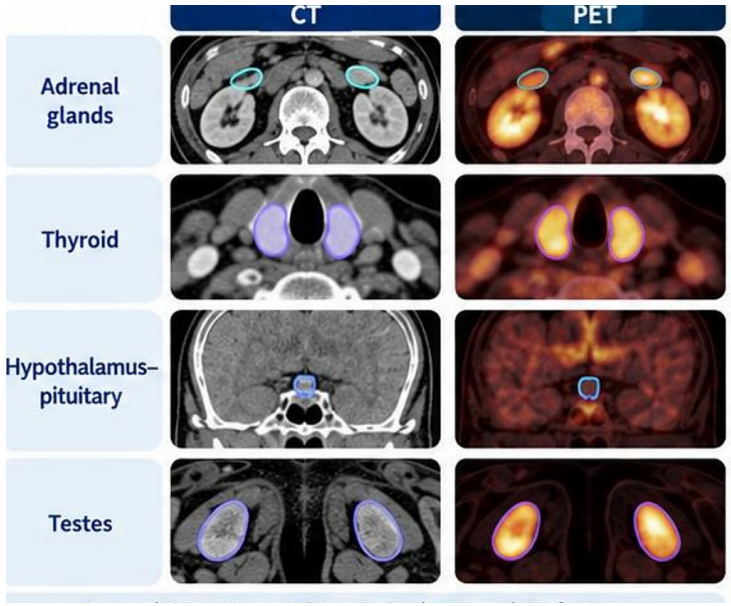
PART I · RESEARCH AND METHODS

Selected studies

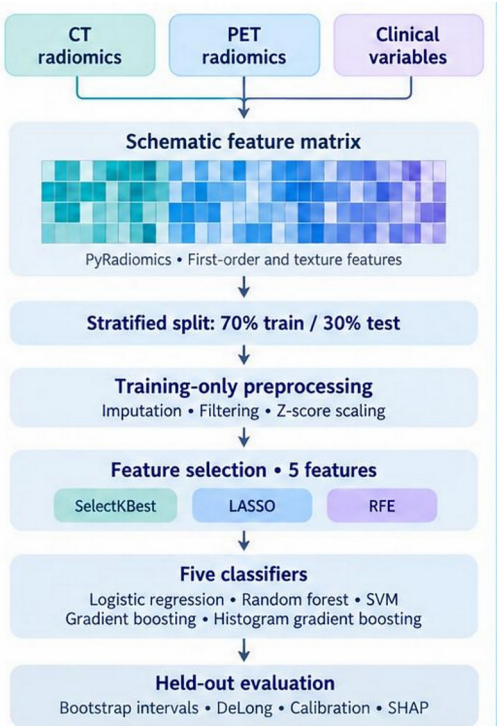
Radiomics · reliability · harmonization · digital twins · predictive dosimetry

RADIOMICS · SYSTEMIC PHENOTYPING

Endocrine-organ PET/CT radiomics



Endocrine organs on PET/CT



CT + PET + clinical fusion model

Results

Test AUC (held-out 30%)

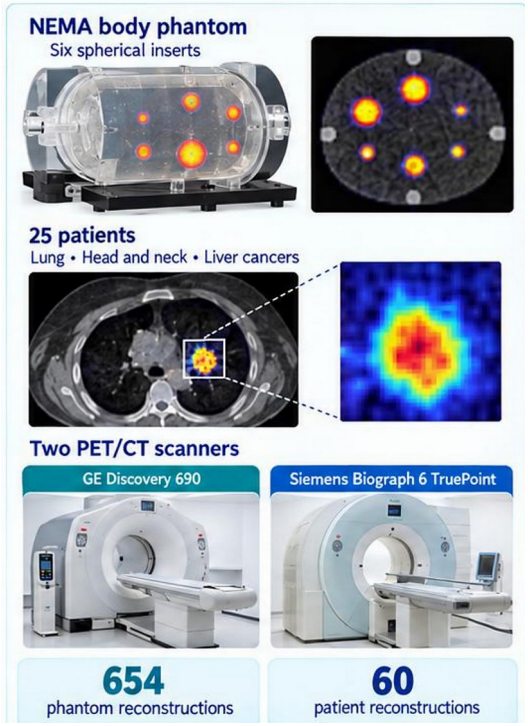
Model	AUC
Clinical only	0.727
CT + clinical	0.729
PET + clinical	0.733
CT + PET + clinical	0.758
95% CI (best)	0.653–0.849
ΔAUC vs clinical	0.031
DeLong p	0.569

Retrospective, single centre

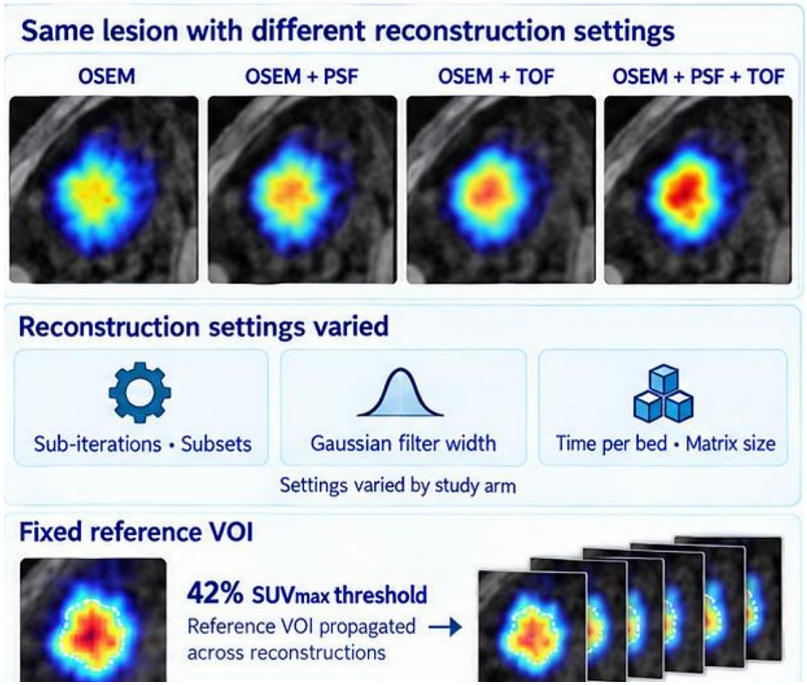
Systemic imaging signatures beyond visible lesions

RADIOMICS · ROBUSTNESS

PET reconstruction changes radiomic features



Phantom + patients, 2 scanners



Same lesion, many reconstructions

Results

Feature variability (COV)

Measure	Features
COV ≤5% (robust)	47%
COV >20% (unstable)	10%
Geometry, COV ≤5%	100%
Intensity, COV ≤5%	44%
Texture, COV ≤5%	41%
Matrix size, COV >20%	56%
Matrix size, COV ≤5%	6%

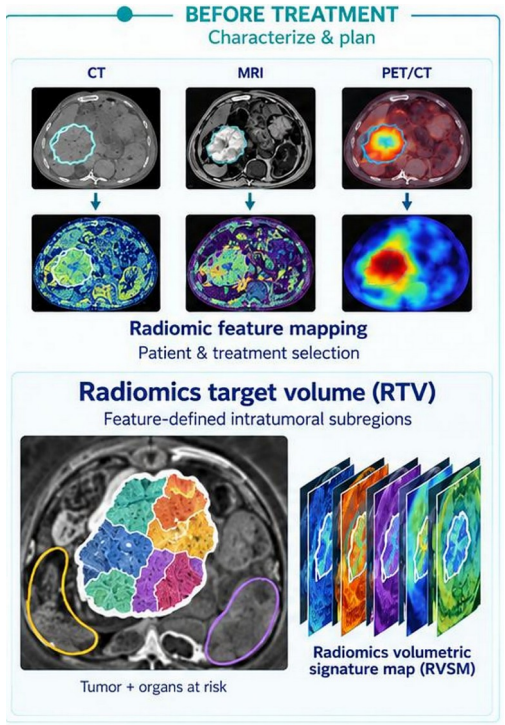
654 phantom + 60 patient reconstructions

Standardize reconstruction before using radiomics

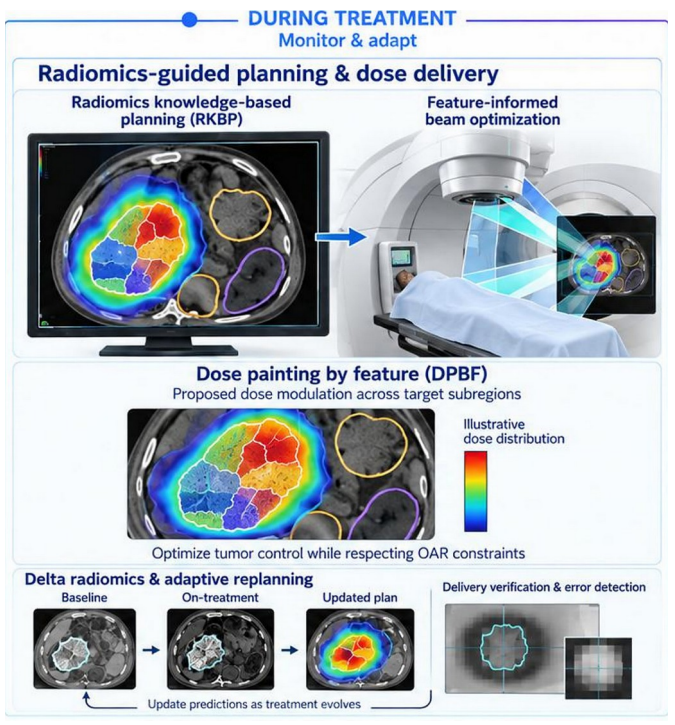
Shiri et al., Eur Radiol 2017

RADIOMICS-GUIDED RADIATION THERAPY

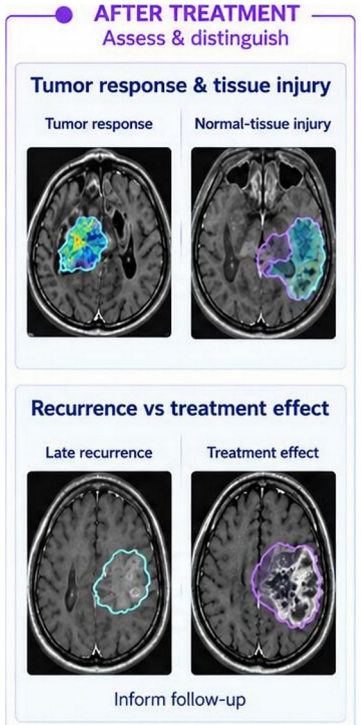
Radiomics across the radiotherapy timeline



Before: characterize and plan



During: monitor and adapt

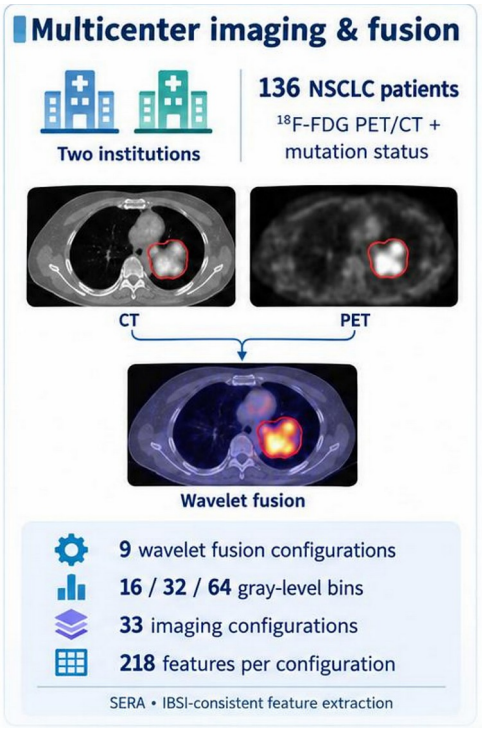


After: assess response

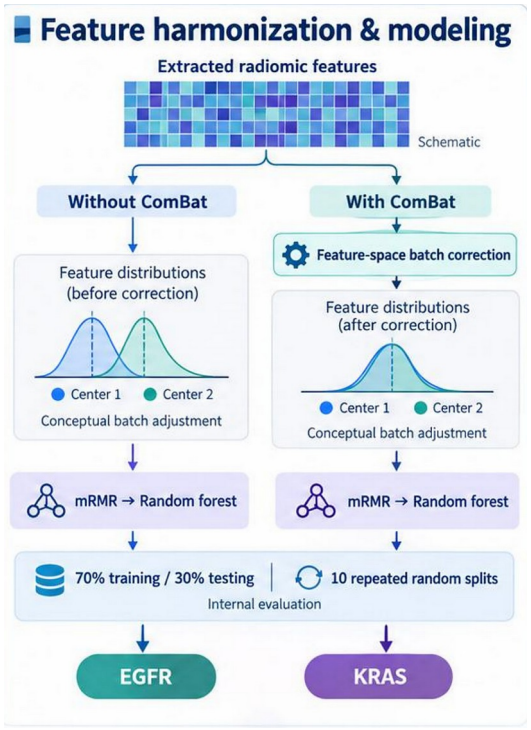
From feature maps to dose painting and adaptive replanning

RADIOGENOMICS · MULTICENTRE HARMONIZATION

ComBat harmonization for EGFR and KRAS



136 NSCLC, two centres



Models with and without ComBat

Results

Mean model AUC

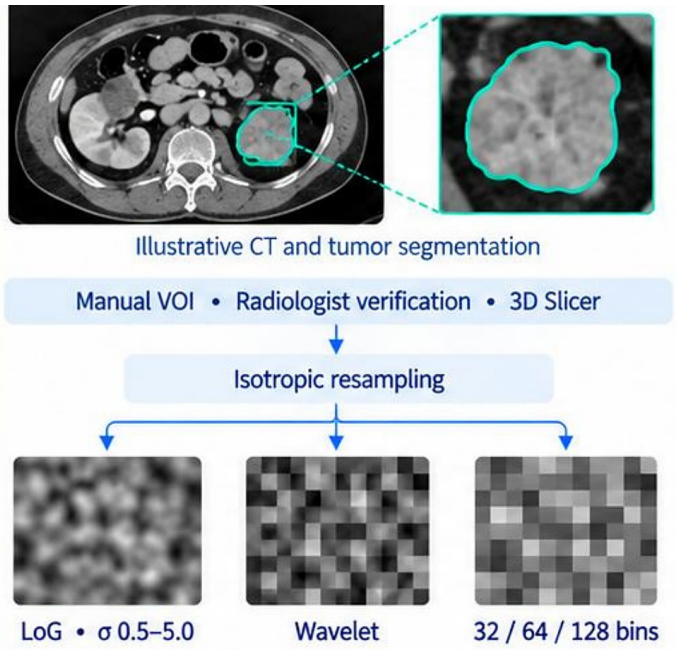
Gene	Before	After
EGFR	0.87–0.90	0.92–0.94
KRAS	0.85–0.90	0.91–0.94
EGFR, best	—	0.94
KRAS, best	—	0.93

Multivariate improvement $q < 0.05$

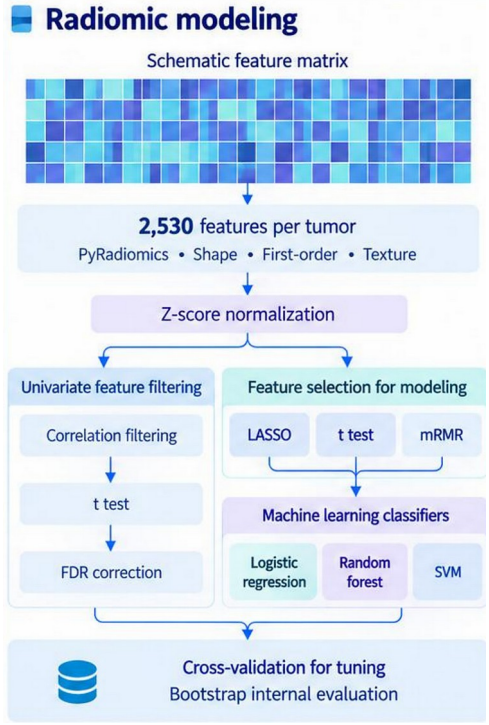
Harmonization makes multicentre radiomics more reliable

RADIOMICS · TUMOUR GRADING

CT radiomics for Fuhrman grade in ccRCC



71 ccRCC CT scans, filtered images



2,530 features → machine learning

Results

AUC by preprocessing

Images	LR	RF	SVM
Original	0.68	0.62	0.76
32 bins	0.73	0.56	0.70
64 bins	0.70	0.55	0.65
128 bins	0.75	0.60	0.77
LoG σ 4.5	0.62	0.62	0.83

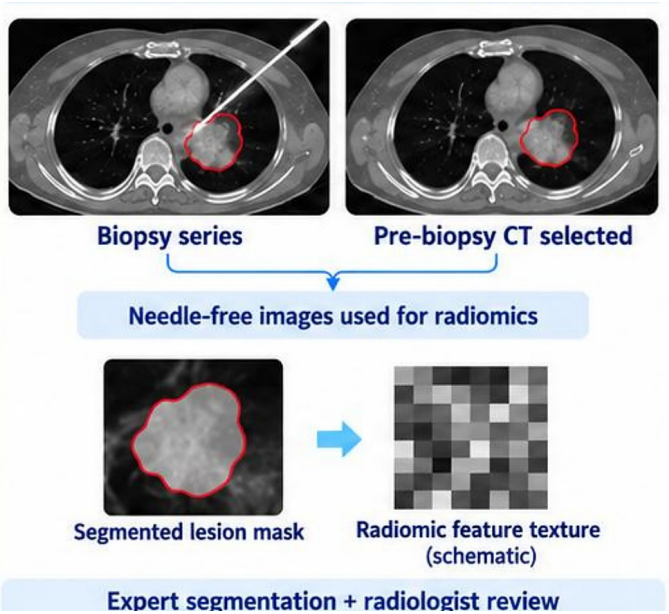
4 texture features significant (FDR q < 0.05)

Potential non-invasive preoperative grading

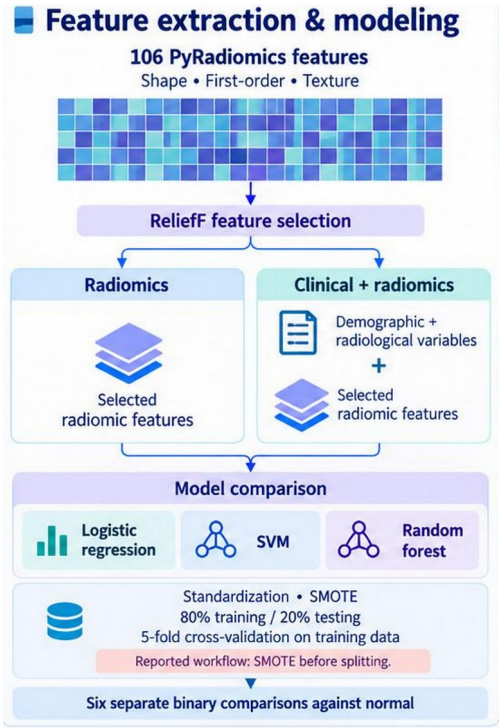
Nazari et al., La Radiologia Medica 2020

RADIOMICS · VIRTUAL BIOPSY

CT-based virtual biopsy of thoracic lesions



Pre-biopsy CT, 300 patients



Feature selection + three classifiers

Results

Test AUC vs normal

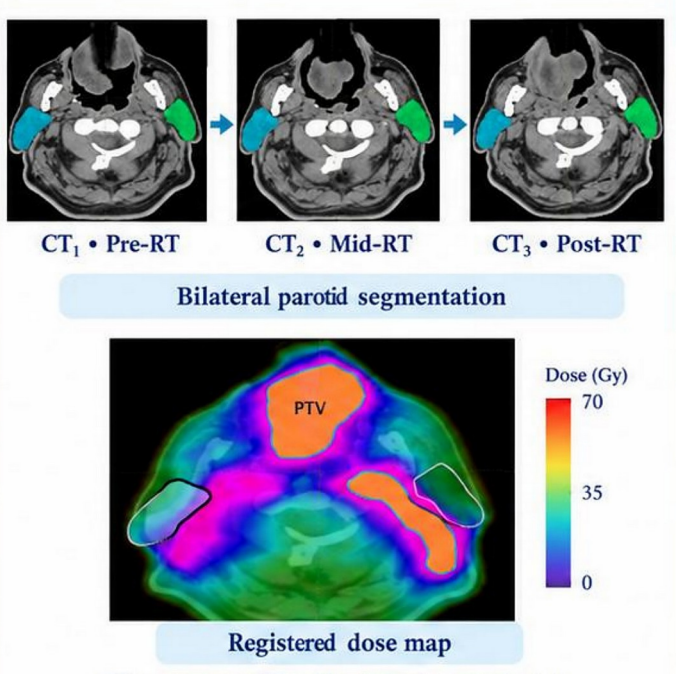
Lesion	Radiom.	+ Clin.
Tumoral	0.682	0.721
Adenocarcinoma	0.690	0.741
SCC	0.820	0.769
SCLC	0.869	0.875
Lymphoma	0.875	0.920
Mesenchymal	0.861	0.852

SMOTE before splitting may inflate results

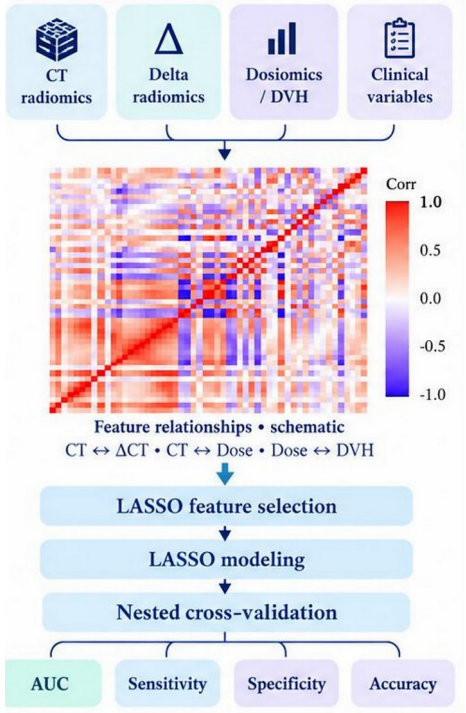
A virtual biopsy alongside tissue diagnosis

RADIOMICS · TOXICITY PREDICTION

Delta radiomics and dose for late xerostomia



Parotid CT before, during, after RT



CT, delta, dose and clinical features

Results

AUC, nested CV

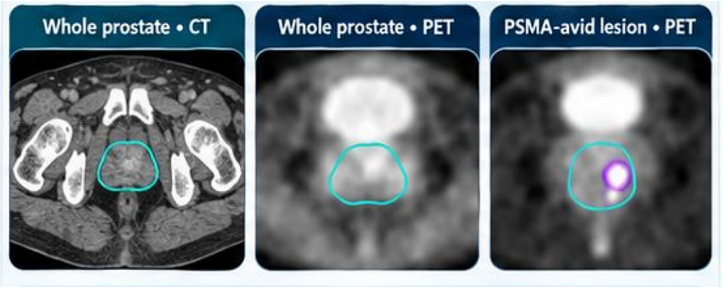
Model	AUC
CT1 (pre-RT)	0.68
CT2 (mid-RT)	0.84
CT3 (post-RT)	0.89
ΔCT2−1 + dose	0.89
DVH	0.89
Dose	0.72
Clinical	0.88

31 patients, 59 parotids

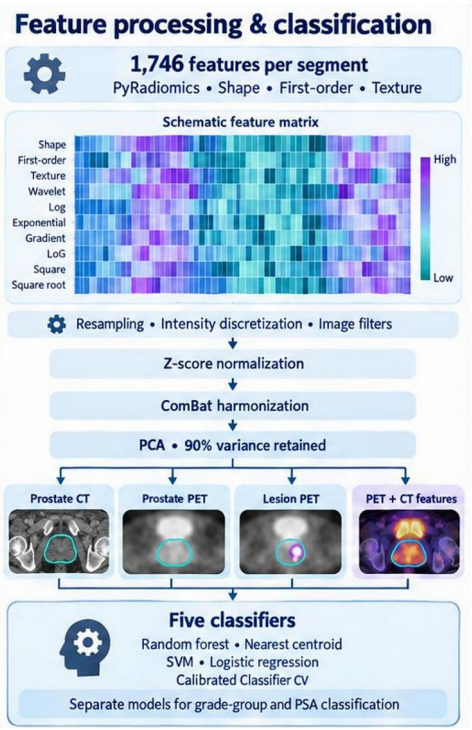
Longitudinal imaging and dose predict dry mouth

RADIOMICS · PSMA PET/CT

Multicentre PSMA PET/CT radiomics for risk



PSMA PET/CT, 243 patients, 4 centres



Harmonized features, five classifiers

Results

Mean accuracy

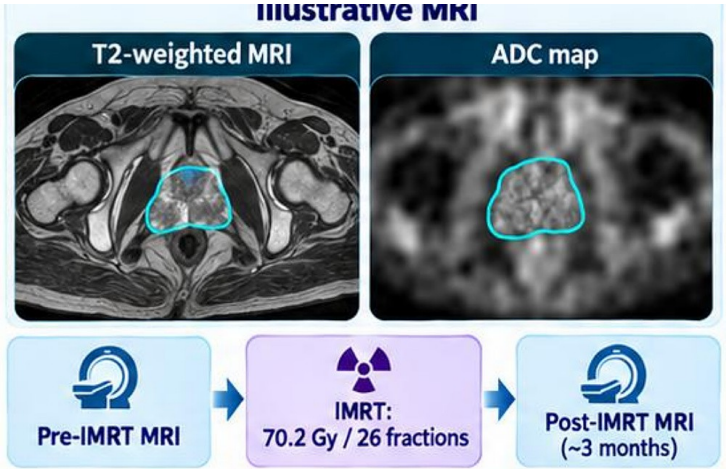
Task	Internal	External
GGG 1–3 vs 4–5	78.0%	73.8%
PSA <20 vs ≥20	75.4%	76.4%

186 development, 57 external test; classifier differences p > 0.05

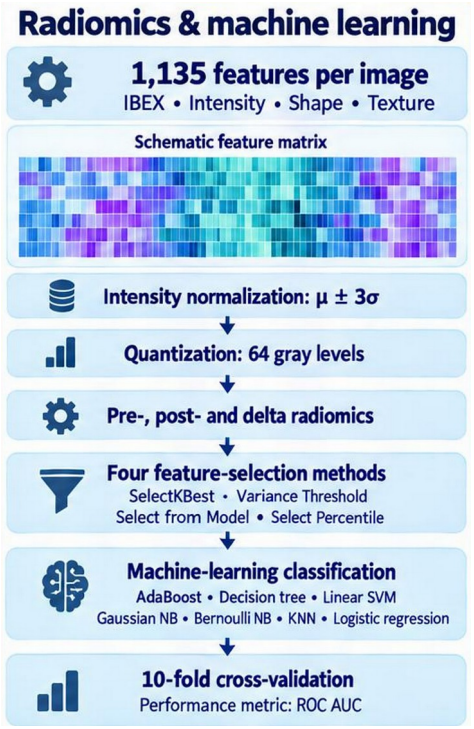
Non-invasive risk stratification from PSMA PET/CT

RADIOMICS · TREATMENT RESPONSE

MRI radiomics for IMRT response in prostate



T2 MRI and ADC before and after IMRT



1,135 features, seven classifiers

Results

AUC

Task	T2w	ADC
IMRT response	0.78	0.77
Gleason	0.739	0.700
Stage	0.625	0.675

Response: best model; Gleason, stage: mean AUC. Responder = ADC rise >20%

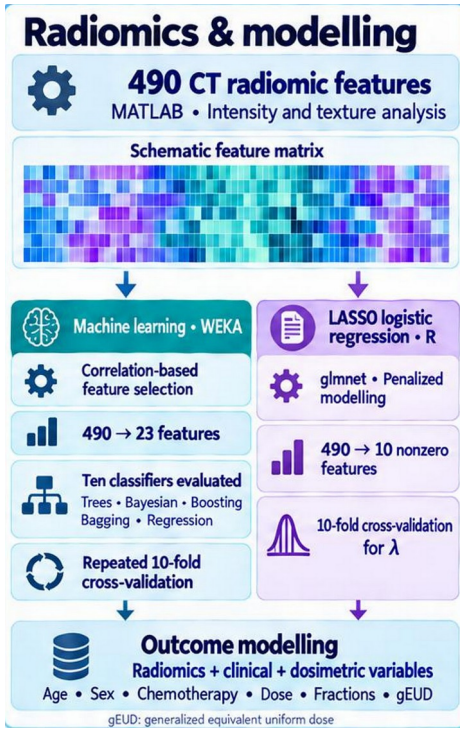
MRI radiomics for response and tumour characterization

COCHLEAR CT RADIOMICS · TOXICITY

Predicting radiation-induced hearing loss



Cochlear CT, 47 patients



490 features, ML and LASSO

Results

Hearing-loss prediction

Model	Result
LASSO radiomics, AUC	0.885
ML accuracy	76.1%
ML precision	75.9%
gEUD only, AUC	0.79
gEUD only, accuracy	72%

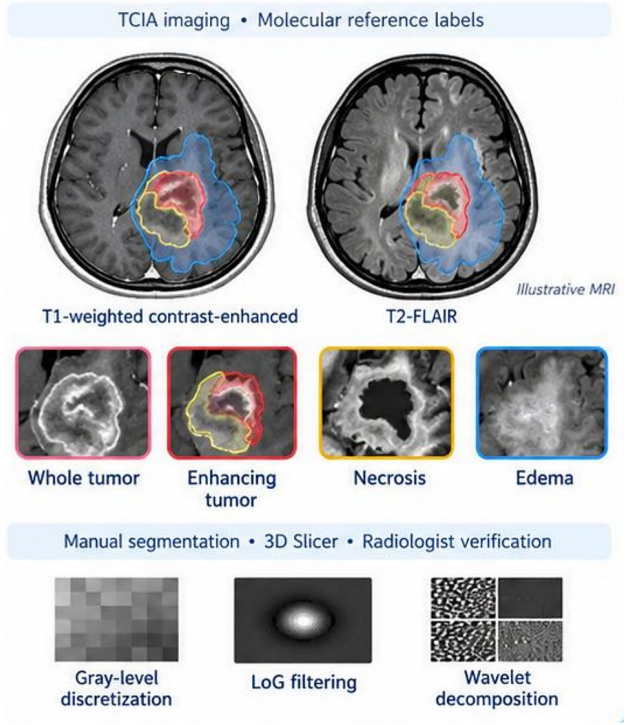
37 of 94 ears with SNHL

Imaging the cochlea to predict hearing loss

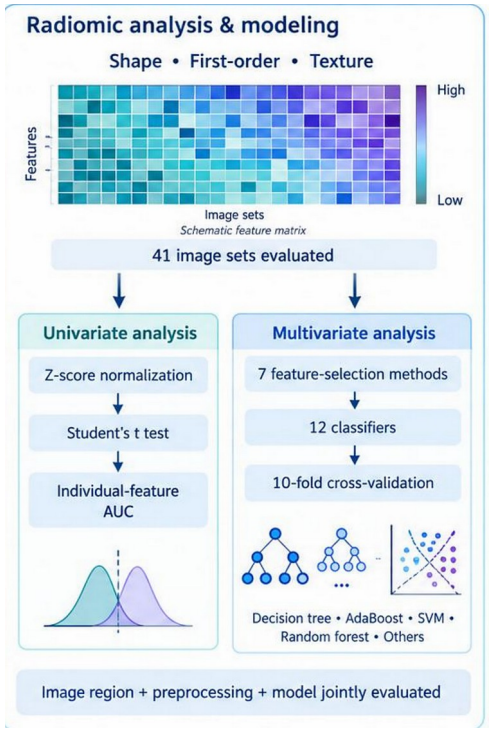
Abdollahi et al., Physica Medica · doi:10.1016/j.ejmp.2017.10.008

RADIOGENOMICS · GLIOBLASTOMA

MRI radiomics for MGMT promoter methylation



MRI tumour sub-regions, 82 patients



41 image sets, 12 classifiers

Results

AUC, 10-fold CV

Model	AUC
Decision tree, edema	0.78
AdaBoost, edema	0.74
Best single feature (GLCM)	0.71

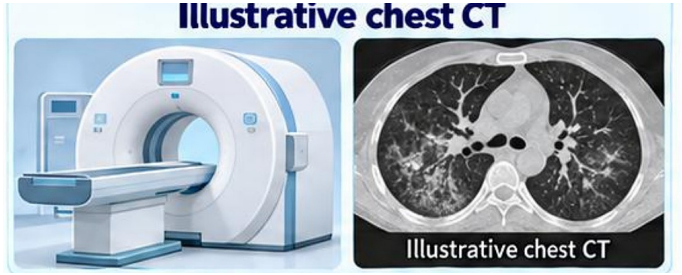
45 methylated, 37 unmethylated

Non-invasive MGMT status from MRI

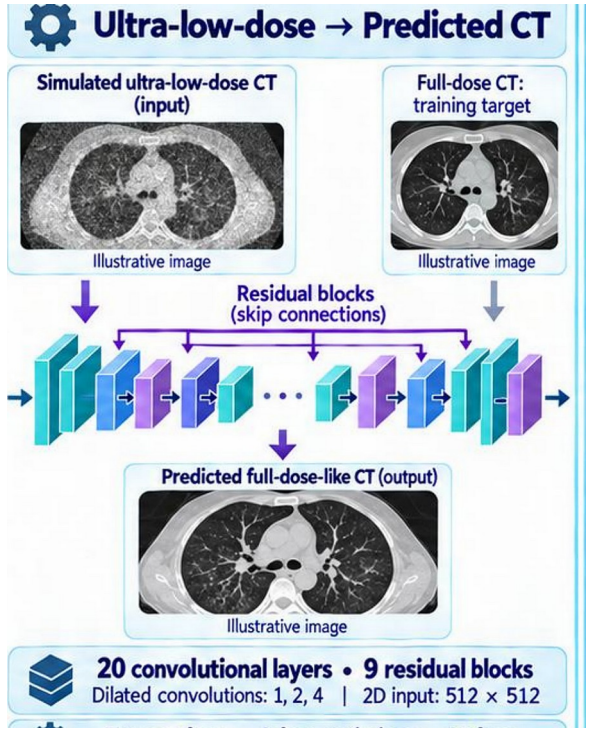
Hajianfar et al., World Neurosurgery 2019

DEEP LEARNING · IMAGE QUALITY

Deep learning for ultra-low-dose chest CT



1,141 chest CTs, 9 centres



Residual network: low → full dose

Results

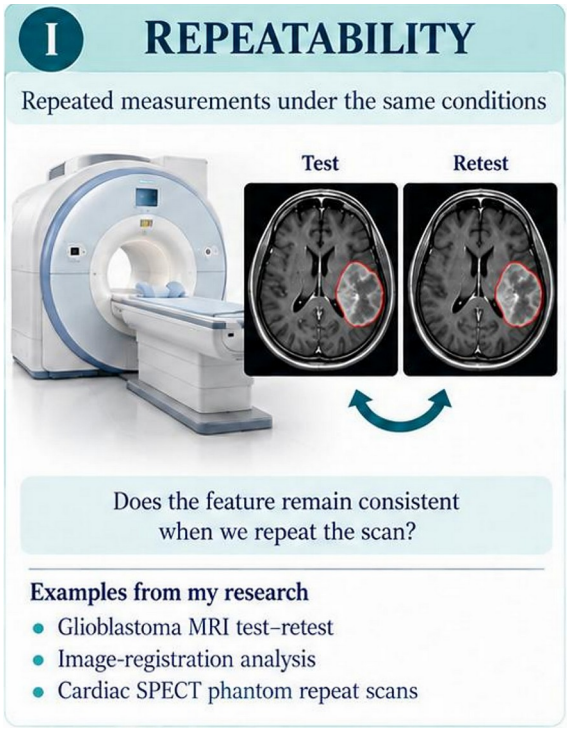
External validation

Metric	ULD CT	DL CT
RMSE	0.16	0.08
SSIM	0.89	0.97
PSNR (dB)	29.40	33.60
Reader score /5	2.78	4.42

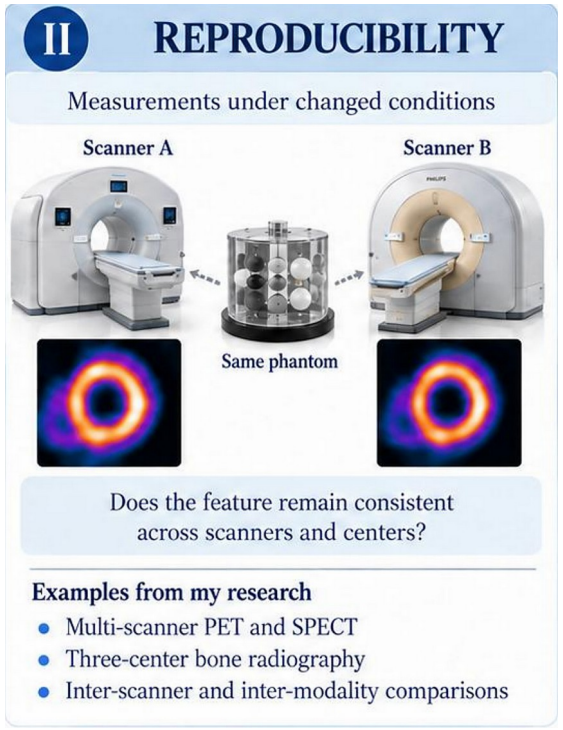
Full-dose reader score 4.72; all $p < 0.0001$

Lower dose with preserved image quality

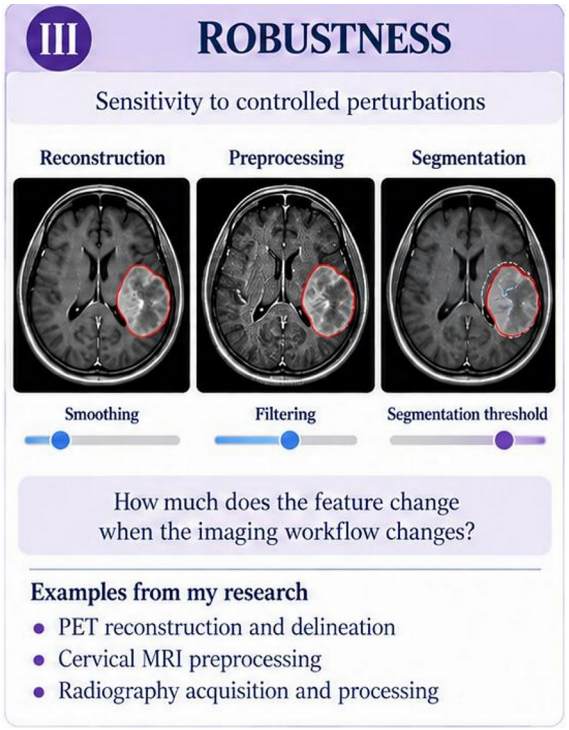
Repeatability, reproducibility and robustness



Repeatability: same conditions



Reproducibility: across scanners

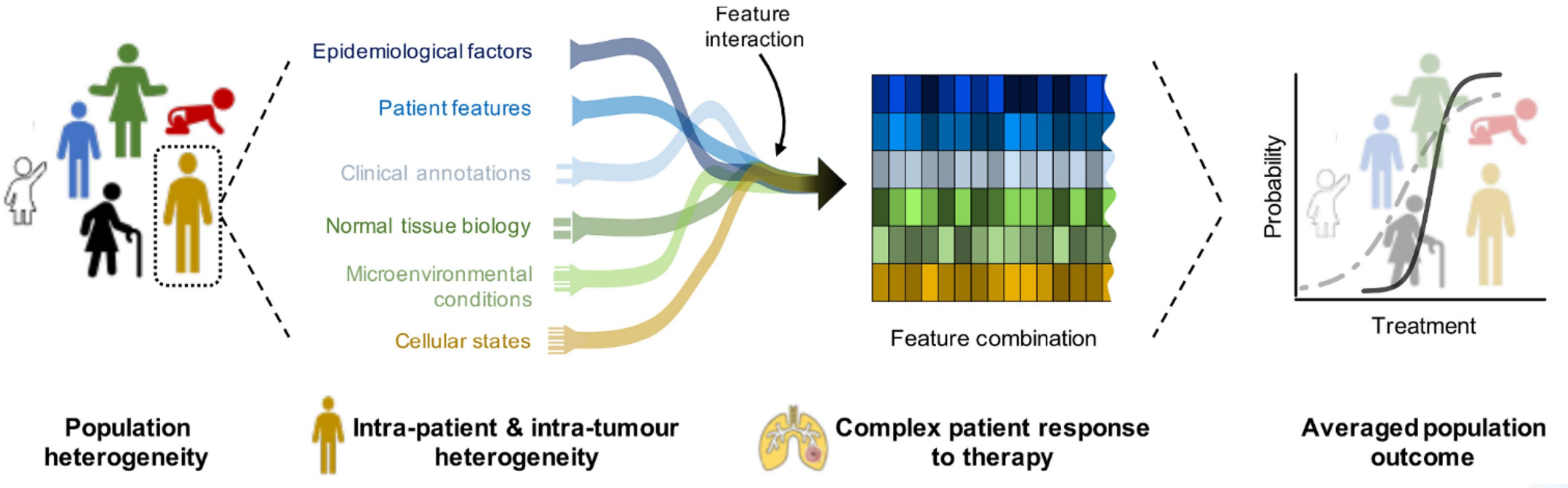


Robustness: workflow changes

Goal: find stable features and the sources of variability

THE CHALLENGE

Treatment is designed for the average patient



Averaging hides the individual patient

THE SOLUTION · DIGITAL TWINNING

Trials for patients who are not average



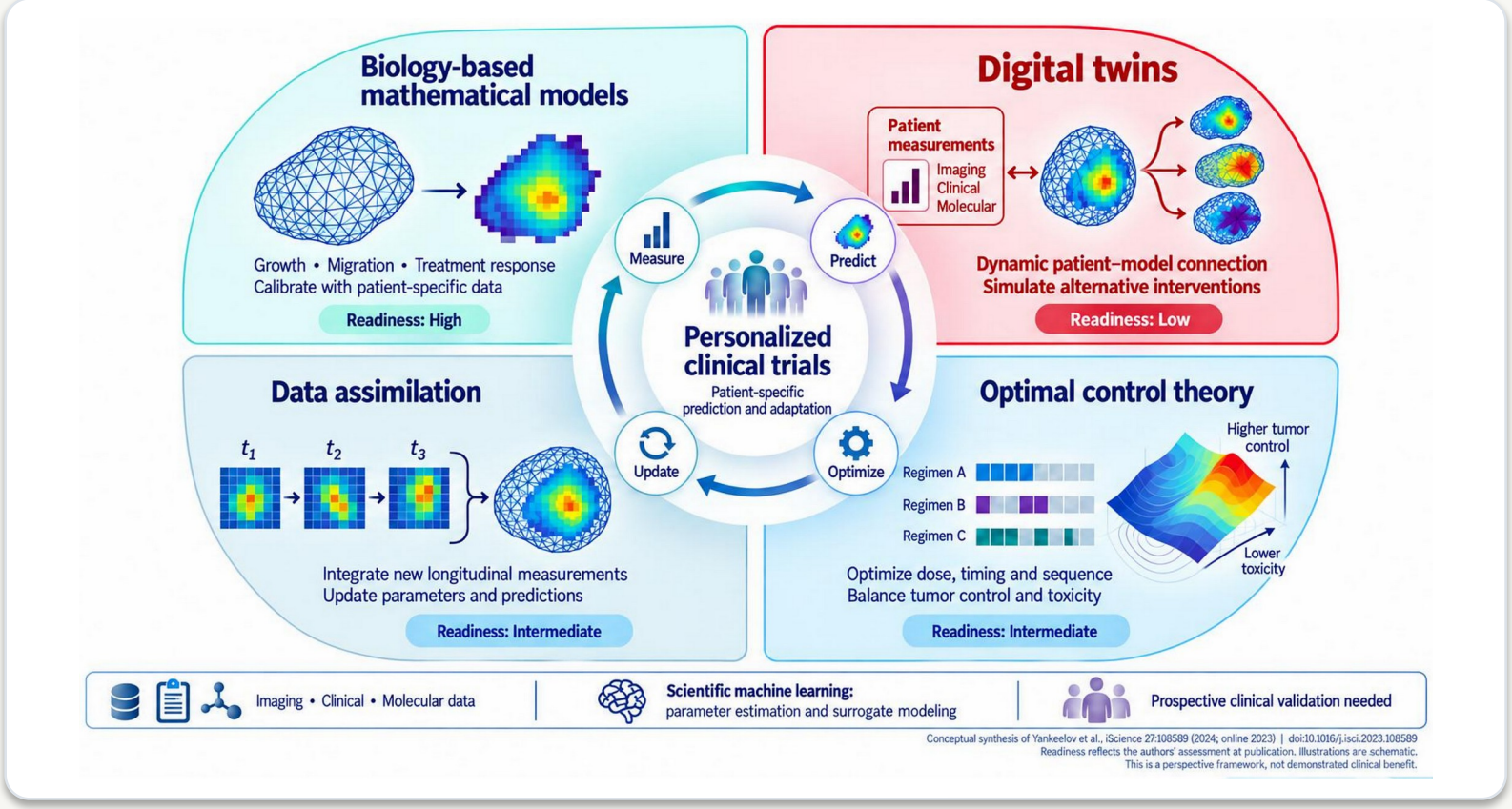
The idea

- Model each patient, not the average
- Calibrate with their own data
- Update as treatment proceeds

From averaged evidence to personalized, adaptive decisions

THE SOLUTION · DIGITAL TWINNING

Four building blocks for personalized trials



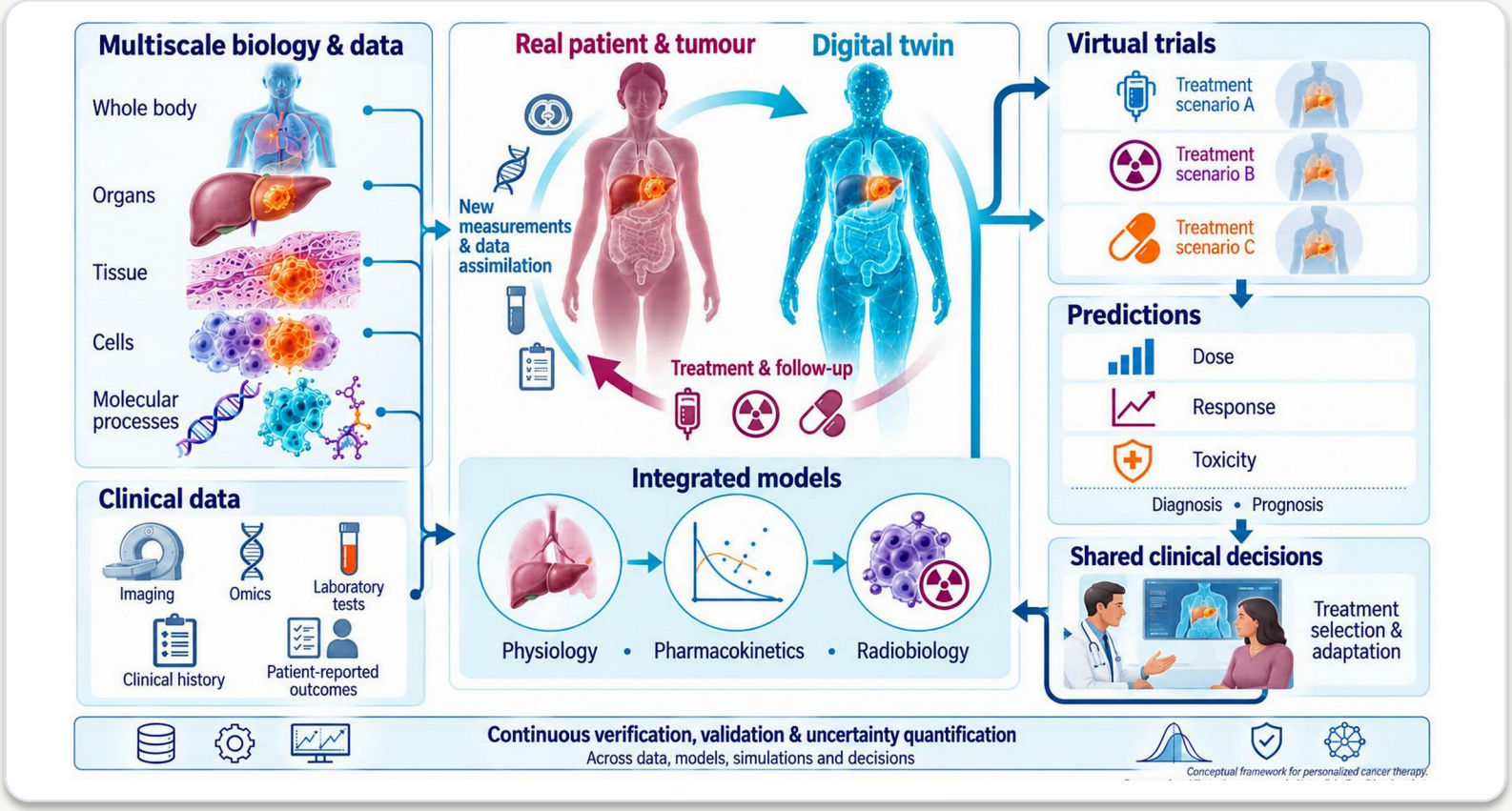
Building blocks

- Biology-based models
- Data assimilation
- Optimal control
- Digital twins

Measure → predict → optimize → update

THE SOLUTION · DIGITAL TWINNING

What is a digital twin?



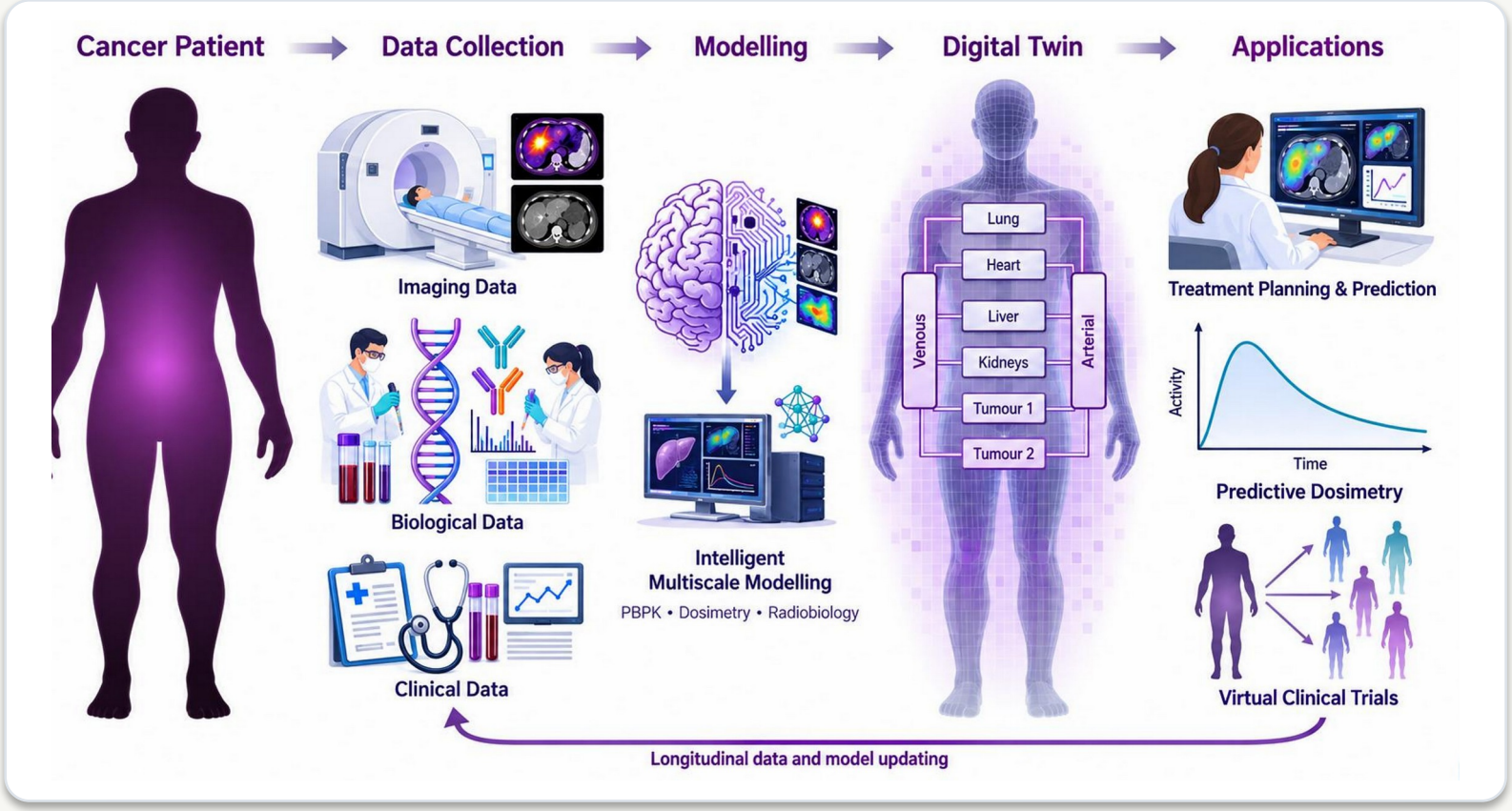
Definition

- A virtual copy of one patient
- Updated with that patient's data
- Predicts and guides decisions

A two-way link between patient and model

THERANOSTIC DIGITAL TWINNING

From the patient to a theranostic digital twin



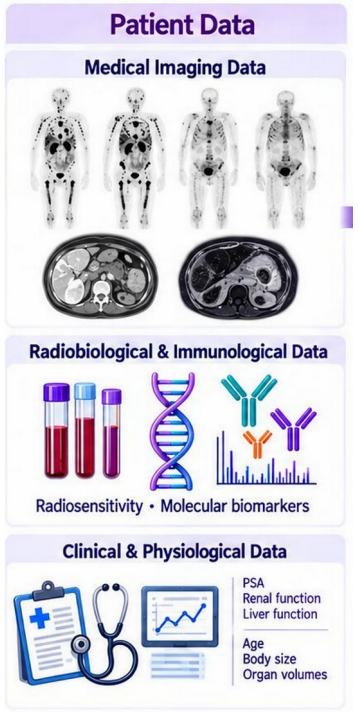
Steps

- Imaging, biological, clinical data
- PBPK, dosimetry, radiobiology
- Planning and virtual trials

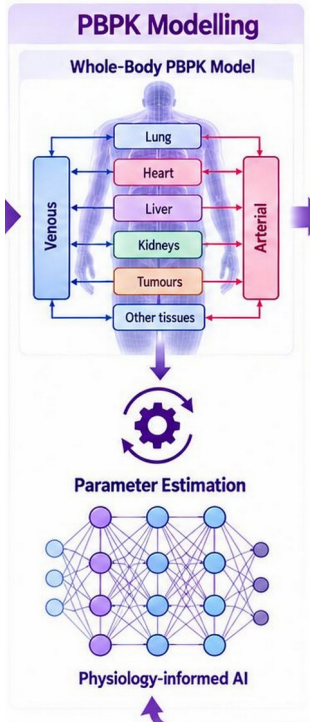
Longitudinal data keep updating the twin

THERANOSTIC DIGITAL TWINNING

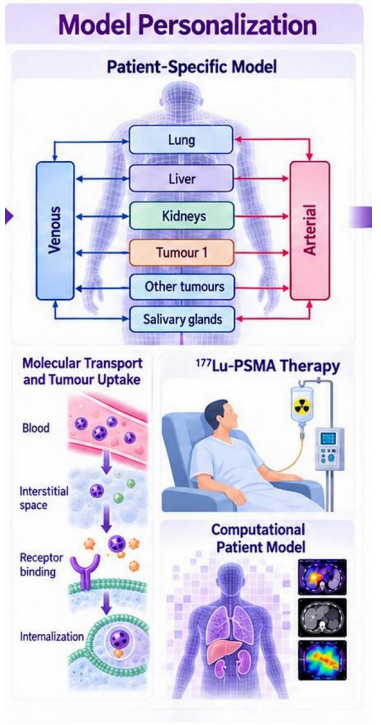
Building the twin: PBPK model for ^{177}Lu-PSMA



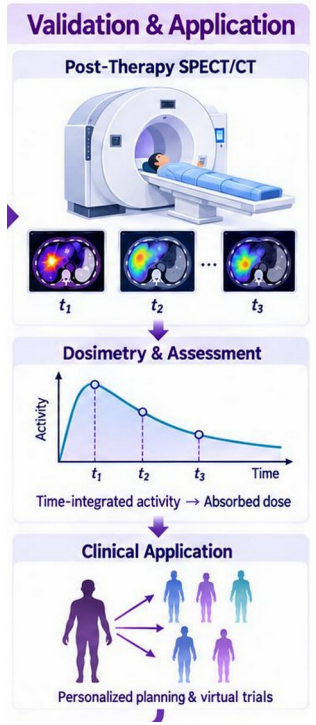
Patient data



PBPK model



Personalization

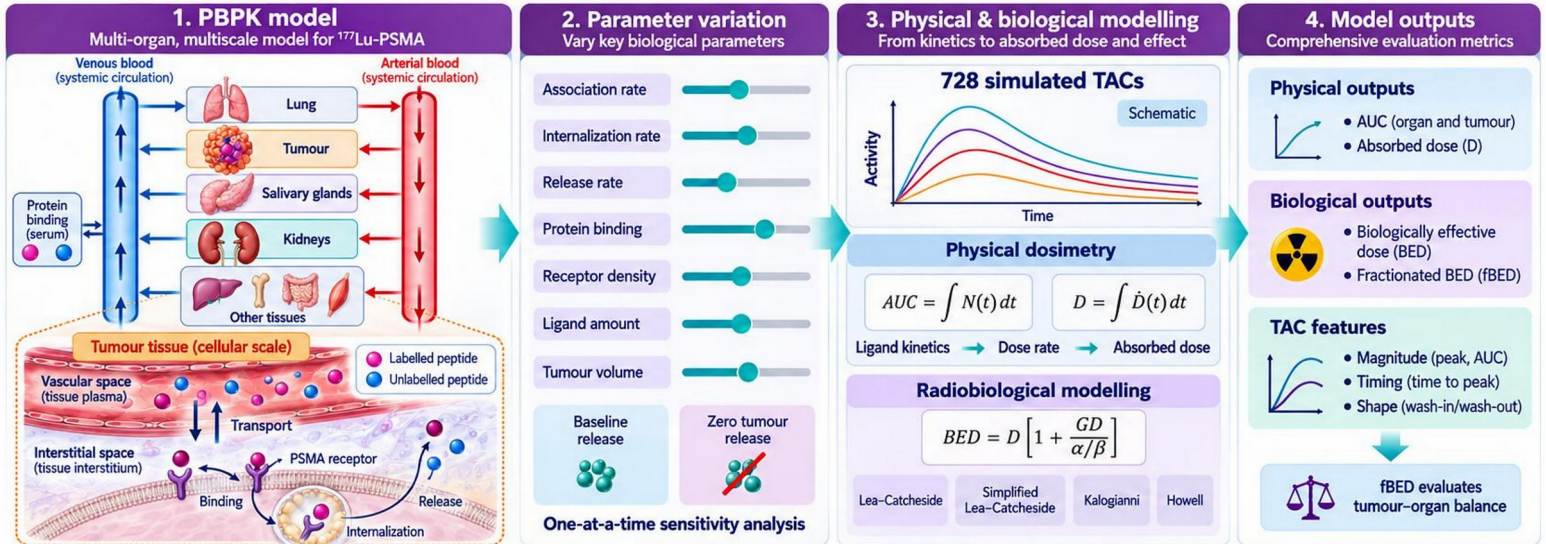


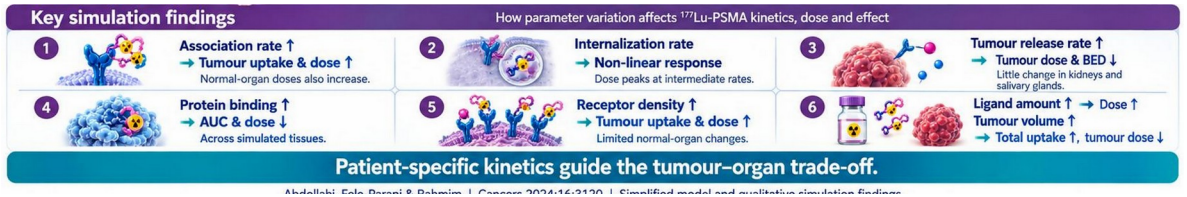
Validation & use

Model refinement with longitudinal data closes the loop

THERANOSTIC DIGITAL TWINNING

Model-informed ^{177}Lu-PSMA optimization

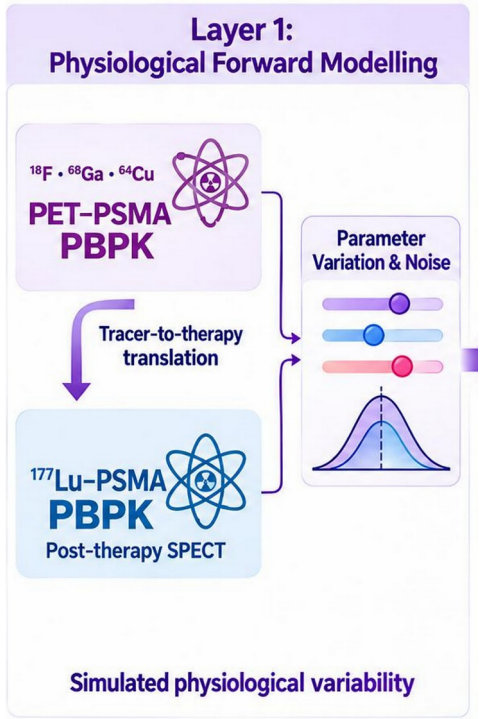




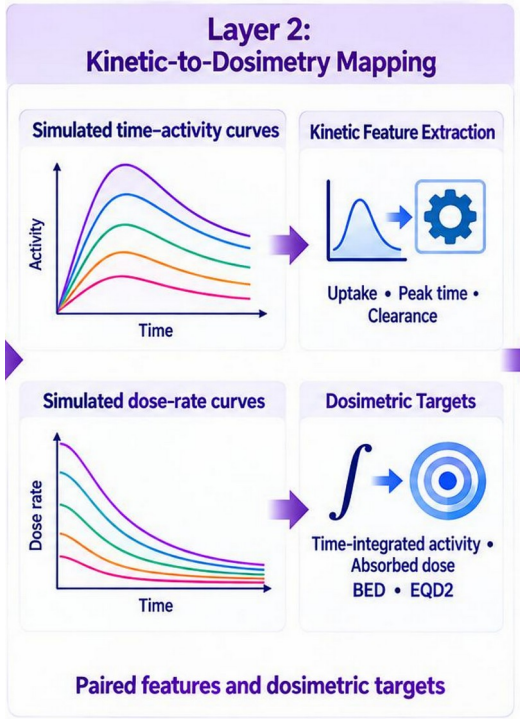
Patient-specific kinetics guide the tumour–organ trade-off

APPLICATION · PREDICTIVE DOSIMETRY

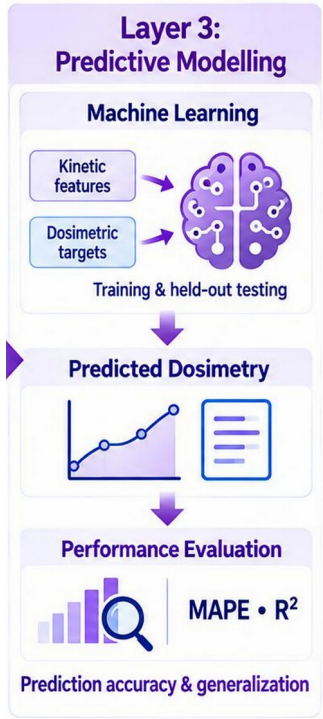
Predicting ^{177}Lu dosimetry from PET kinetics



Layer 1: physiological modelling



Layer 2: kinetics → dosimetry



Layer 3: machine learning

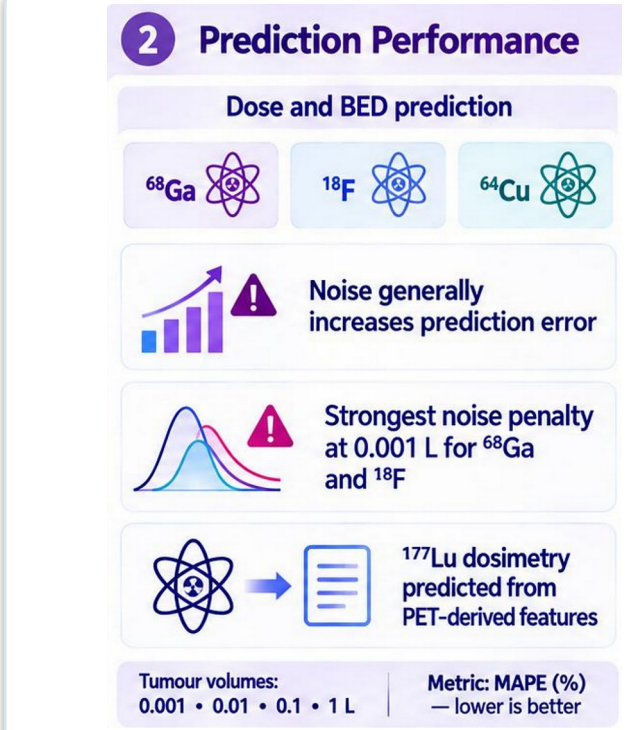
Pre-therapy PET kinetics → predicted therapy dose

APPLICATION · PREDICTIVE DOSIMETRY

Predictive dosimetry: key findings



Peak uptake (AMax) leads



Noise matters most in small tumours



Model interpretation (SHAP)

Uptake magnitude and kinetics support dosimetry prediction

Virtual cohort and simulated dosimetry

Study design

Item	Value
Virtual patients	640
TACs simulated	15,360
PET tracers	^{18}F, ^{68}Ga, ^{64}Cu
Therapy	^{177}Lu-PSMA
Regions	Tumour + 5 organs
Tumour volumes	0.001–1 L
Validation	5-fold CV
Primary endpoint	MAPE

PBPK-simulated ^{177}Lu dosimetry

Mean ± SD across virtual patients

Quantity	Small tumour	1 L tumour
Tumour dose (Gy)	48.3 ± 38.1 (0.01 L)	31.6 ± 20.4
Tumour BED (Gy)	58.4 ± 57.0	36.5 ± 26.7
Tumour EQD2 (Gy)	40.6 ± 38.8	24.1 ± 17.6
Tumour AUC (×10^6 nmol)	0.03 ± 0.03	21.6 ± 14.0
Kidney / tumour ratio	27.7	0.029
Kidney dose (Gy)	≈ 4.7 ± 4.1	—
Spleen, marrow dose	< 0.1 Gy	< 0.1 Gy

AMax vs AUC: r = 0.996 · AMax vs dose, BED, EQD2: r > 0.96

Larger tumours act as a sink: tumour exposure rises while organ doses stay low

Prediction accuracy from pre-therapy PET

Tumour prediction

MAPE with PET-like noise · smallest (0.001 L) → largest (1 L) tumour

Tracer	Tumour AUC	Tumour dose
^{64}Cu-PSMA-617	**~5% → ~2%**	**~12% → 8%**
^{18}F-DCFPyL	>6% → ~2–3%	>20% → ~12%
^{68}Ga-PSMA-11	flat, broad	wide variability

- Error falls as tumour volume rises
- Tumours ≥ 0.1 L: predictions track the identity line
- Ridge and XGBoost most robust to noise

Normal-organ dose

MAPE (%) with noise · tumour volume 0.1 L

Organ	^{18}F	^{68}Ga	^{64}Cu
Kidney	16.3	16.9	**11.3**
Salivary glands	27.2	20.8	**15.8**
Liver	28.0	30.6	**14.4**
Spleen	28.6	36.8	29.7
Red marrow	19.2	25.4	**10.4**

Lowest error per organ highlighted. Without noise, errors were roughly halved.

^{64}Cu-PSMA-617 PET gave the most robust predictions: tumour dose MAPE down to 8%, organs 10–20%

PART II · 10 MINUTES

Research vision

Five aims: imaging biomarkers → AI imaging → digital twins → advanced therapy → virtual trials

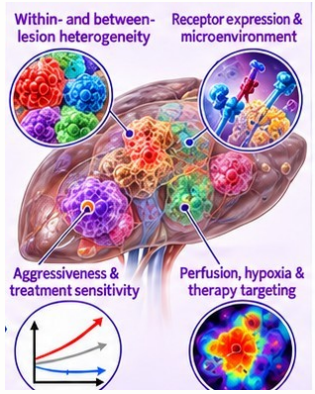
AIM I

Quantitative molecular imaging biomarkers



1 Measure disease

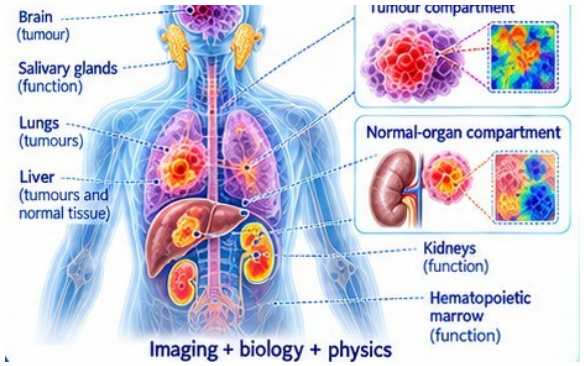
- Combine PET/CT, SPECT and MRI
- Burden, target expression, tracer kinetics
- Change over time; PSMA and SSTR theranostics





2 Characterize biology

- Heterogeneity within and between lesions
- Receptors, perfusion, hypoxia; normal-organ uptake and clearance
- Radiomics/AI + molecular and clinical data → stratification and targeting



3 Predict and personalize

- Digital twin of tumour + normal organs
- Predictive dosimetry: tumour control vs toxicity
- Virtual trials; models updated with new imaging

Goal: a predictive, therapy-guiding imaging platform — reproducible and clinically validated.

Concept adapted from Tharmaseelan et al., Cancers 2022;14:3349 (CC BY 4.0), extended to the proposed research vision

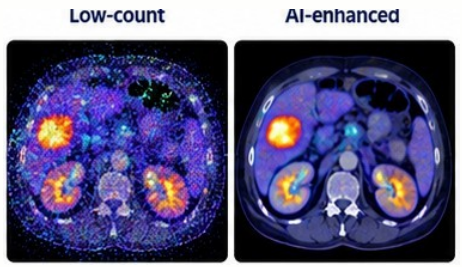
AIM II

AI-enhanced molecular imaging



1 Acquire

- Physics-informed reconstruction
- Lower-dose, faster PET/SPECT
- Quantitative accuracy kept



2 Enhance

- AI denoising
- Lesion detection and segmentation
- Registration over time



3 Quantify

- Disease burden and heterogeneity
- Radiomics + clinical + biology
- Personalized dosimetry



4 Support

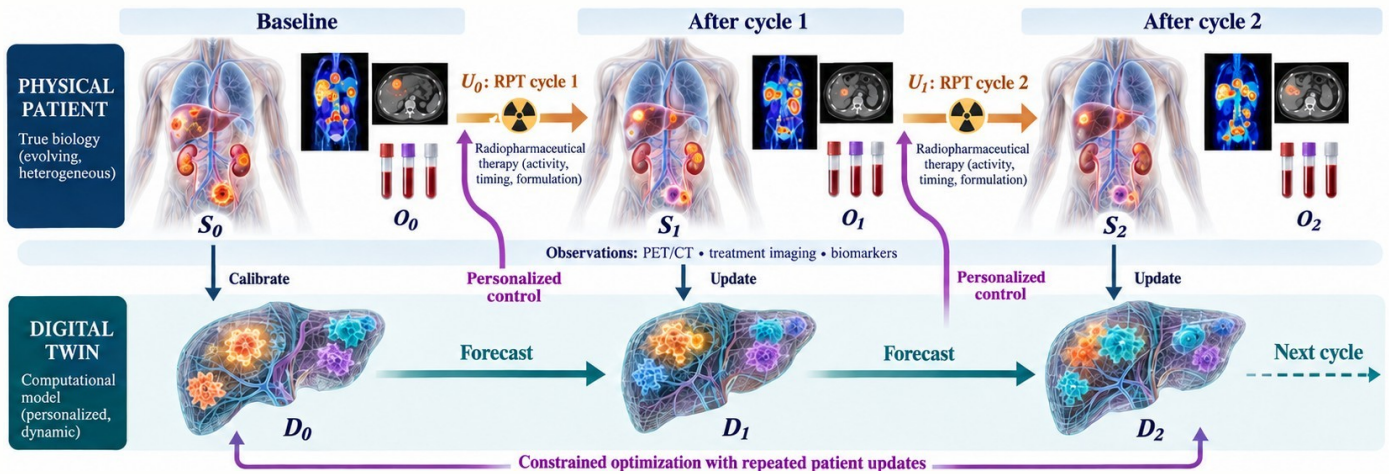
- Therapy-response prediction
- Patient stratification
- Interpretable decision support

Priorities: physical consistency · biological interpretability · robust validation — less repetitive work for clinicians.

Proposed research framework · informed by Herskovits (2021) and Mansouri et al. (2026)

AIM III

Digital twins for radiopharmaceutical therapy



Imaging and biomarkers calibrate the twin at baseline and update it after every cycle; the twin forecasts and sends back personalized control.

1 Mechanistic core

Pharmacokinetics → dose and dose rate → tumour growth, damage, repair, resistance

2 AI supports the model

Patient-specific parameters from imaging; fast surrogate simulations

3 Predict → optimize → treat

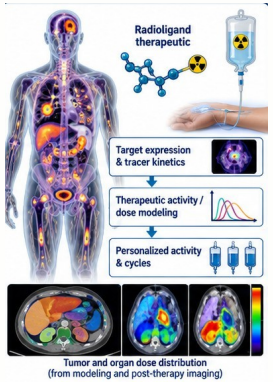
Predict response and toxicity → best activity, timing and number of cycles

Feedback loop: measure → predict → optimize → update — then validate prospectively.

Proposed RPT adaptation of Lorenzo et al., Annu Rev Biomed Eng 2024;26:529–560 (Figs. 4–6)

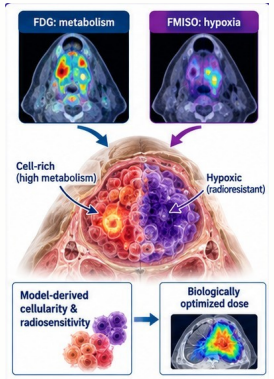
AIM IV

Advanced therapeutic technologies



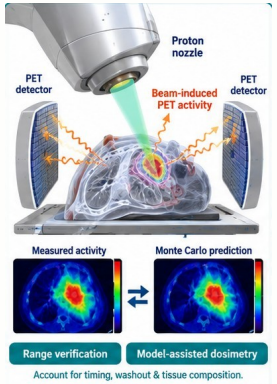
RPT planning

- PET target expression and kinetics
- Personalized activity and cycles
- Checked by post-therapy dosimetry



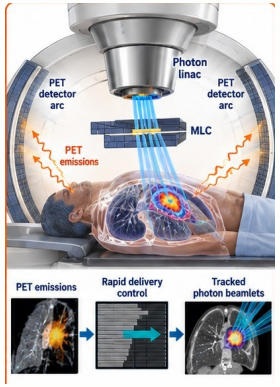
Dose painting

- FDG metabolism + FMISO hypoxia
- Map cellularity and radiosensitivity
- Respect normal-tissue limits



Proton therapy

- Beam-induced PET for range checks
- Model-assisted dosimetry
- PET activity ≠ absorbed dose



Biology-guided RT

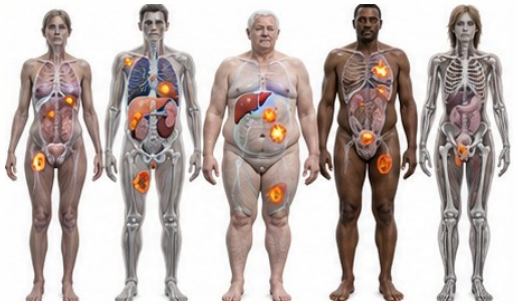
- Fast PET signals guide the beam
- Follows moving targets
- Within a planned dose envelope

Shared goal: plan, verify, measure response and adapt — including RPT + external-beam combinations.

Oderinde et al. (2021) · Lazzeroni et al. (2026) · Abdel-Wahab & Giammarile (2026)

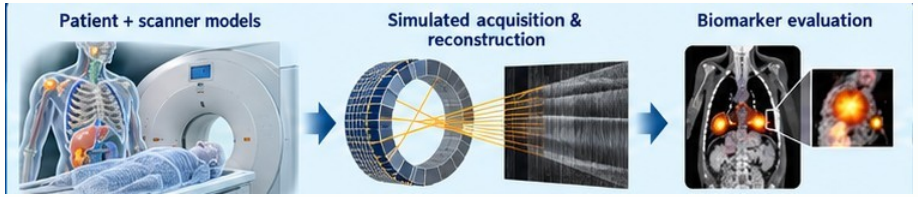
AIM V

Virtual trials and drug development

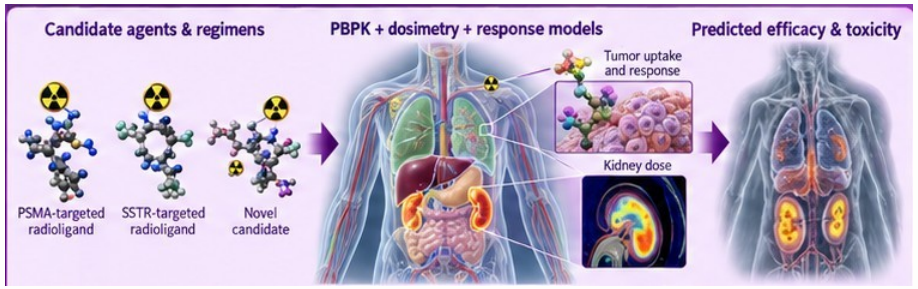


1 Virtual cohorts

- Imaging, biology and clinical data
- Anatomy, burden, kinetics, treatment sensitivity



2 Imaging trials: scanners, protocols, reconstruction; when is dynamic Ki better than static SUV?



3 Therapy trials: PBPK + dosimetry + response for agents, activities and schedules



4 Real-world studies

- Prioritize promising strategies
- Design more informative prospective trials

Credible simulation: physiological plausibility · uncertainty analysis · independent validation

Lindner & Link (2018) · Wangerin et al. (2017) · Kleeberger (2025) · Samei et al. (2025)

COLLABORATION

Working across centres

QURIT lab

BC Cancer & UBC, Vancouver

Theranostic digital twins, PBPK, AI quantification

PIN lab

Geneva University Hospital

AI quantitative imaging, Monte Carlo and deep-learning dosimetry

United Theranostics

International network

Clinical theranostics translation, real-world data

Johns Hopkins University

Baltimore, USA

RPT dosimetry, radiobiology, molecular imaging

Tehran Univ. of Medical Sciences

Nuclear medicine & molecular imaging

Theranostics cohorts, PET/SPECT validation data

Iran Univ. of Medical Sciences

Medical physics & nuclear medicine

Patient dosimetry and AI for ^{177}Lu-PSMA

Bushehr Univ. of Medical Sciences

Persian Gulf NM Research Center

Radionuclide therapy clinic, animal imaging

Thank you

The future is bright — and we are building it together.

Molecular imaging | **Precision medicine** | **Digital twins** | **Virtual trials**

Hamid Abdollahi, PhD · Questions welcome